Problem 4.37: Decays of the η Meson

Griffiths, *Introduction to Elementary Particles*

A Detailed, Pedagogical Solution

Problem Statement (Griffiths 4.37)

The dominant decays of the η meson are:

$$\eta \to 2\gamma \ (39\%), \qquad \eta \to 3\pi \ (55\%), \qquad \eta \to \pi\pi\gamma \ (5\%) \tag{4.78}$$

and it is classified as a 'stable' particle, so evidently none of these is a purely strong interaction. Offhand, this seems odd, since at 549 MeV/c^2, the η has plenty of mass to decay strongly into 2π or 3π.

(a) Explain why the 2π mode is forbidden, for both strong and electromagnetic interactions.

(b) Explain why the 3π mode is forbidden as a *strong* interaction, but allowed as an *electromagnetic* decay.

1 Background: The η Meson Quantum Numbers

Before tackling the problem, let's collect everything we know about the η and the pion.

The η is a pseudoscalar meson in the $I = 0$ sector of the $\ell = 0$, $s = 0$ nonet. From Table 4.6:

Property	**Value**
Mass	549 MeV/c^2
Quark content	$(u\bar{u} + d\bar{d} - 2s\bar{s})/\sqrt{6}$
Isospin I	0
Spin J	0
Orbital angular momentum L	0
Quark spin S	0
Parity $P = (-1)^{L+1}$	-1
C-parity $C = (-1)^{L+S}$	$+1$
G-parity $G = (-1)^I C$	$+1$
J^{PC}	0^{-+}

Property	**Value**
Mass	≈ 140 MeV/c^2
Isospin I	1
Spin J	0
Parity P	-1
C-parity (π^0)	$+1$
G-parity	-1
J^{PC}	0^{-+}

Symmetry	**Strong**	**EM**	**Weak**
Parity P	✓	✓	×
C-parity	✓	✓	×
CP	✓	✓	≈
Isospin I	✓	×	×
G-parity	✓	×	×

G-parity and isospin are only conserved in strong interactions. P, C, and CP are conserved in both strong *and* electromagnetic interactions.

2 Part (a): Why is $\eta \to 2\pi$ Forbidden?

The decay $\eta \to 2\pi$ could in principle occur via two channels: $\eta \to \pi^+\pi^-$ or $\eta \to \pi^0\pi^0$. We will show that *both* are forbidden by *both* strong and electromagnetic interactions, through multiple independent arguments.

2.1 Argument 1: The Parity–Angular Momentum Contradiction

This is the most direct argument.

Step 1: Angular momentum conservation constrains L.
The η has $J = 0$. Both pions have spin-0. Therefore the total angular momentum of the two-pion system is purely orbital:

$$J_{\text{final}} = L$$

Angular momentum conservation requires $J_{\text{final}} = J_\eta = 0$, so:

$$\boxed{L = 0}$$

Step 2: Parity conservation constrains L differently.
The parity of the two-pion system is:

$$P_{\text{final}} = P_\pi \cdot P_\pi \cdot (-1)^L = (-1)(-1)(-1)^L = (-1)^L$$

Parity conservation requires $P_{\text{final}} = P_\eta = -1$:

$$(-1)^L = -1 \quad \Longrightarrow \quad L = \text{odd} \quad (L = 1, 3, 5, \ldots)$$

Step 3: The contradiction.
Angular momentum says $L = 0$ (even). Parity says L must be odd. These two requirements are **mutually incompatible**. No value of L can satisfy both.

$$\underbrace{L = 0}_{\text{from } J \text{ conservation}} \qquad \neq \qquad \underbrace{L = \text{odd}}_{\text{from } P \text{ conservation}}$$

The decay $\eta \to 2\pi$ is forbidden by **parity conservation** (combined with angular momentum conservation) for *any* interaction that conserves parity. Since both the strong and electromagnetic interactions conserve parity, **both** forbid this decay.

A common student question: "Why not just let $L = 1$ and satisfy parity?"

The answer is that L is not a free parameter we can choose. It is *determined* by angular momentum conservation. The η has $J = 0$, and both pions have $J = 0$. The *only* angular momentum in the final state is L. So:

$$J_{\text{total}} = L + 0 + 0 = L$$

Conservation of angular momentum forces $L = 0$. Period. We cannot "choose" $L = 1$ — that would violate angular momentum conservation, which is *always* exact.

2.2 Argument 2: CP Violation (Even Stronger)

Even if we temporarily forget about the parity argument, CP conservation independently forbids this decay for *all* values of L.

Step 1: CP of the η.
Since $C(\eta) = +1$ and $P(\eta) = -1$:

$$CP(\eta) = C \times P = (+1)(-1) = -1$$

Step 2: CP of the $\pi^+\pi^-$ system.
For a $\pi^+\pi^-$ pair, the charge conjugation operator swaps $\pi^+ \leftrightarrow \pi^-$. In the center-of-mass frame, this is equivalent to reflecting the relative position vector $\vec{r} \to -\vec{r}$, which gives a factor $(-1)^L$. The parity of the two-pion system is also $(-1)^L$ (as we computed above). Therefore:

$$C(\pi^+\pi^-) = (-1)^L, \qquad P(\pi^+\pi^-) = (-1)^L$$

$$CP(\pi^+\pi^-) = (-1)^L \cdot (-1)^L = (-1)^{2L} = +1 \quad \text{for ALL } L$$

Step 3: The mismatch.

$$CP(\eta) = -1 \qquad \neq \qquad CP(\pi^+\pi^-) = +1$$

This holds **for every value of** L, not just $L = 0$. There is no escape.

CP conservation alone forbids $\eta \to \pi^+\pi^-$ regardless of the orbital angular momentum. Since both strong and electromagnetic interactions conserve CP, this decay is forbidden by both.

2.3 Argument 3: Bose Symmetry Kills $\eta \to \pi^0\pi^0$

For the neutral channel $\eta \to \pi^0\pi^0$, there is a *third* independent argument from quantum statistics.

Two π^0 mesons are **identical bosons** (same mass, same charge, same quantum numbers). The total wavefunction must be **symmetric** under exchange.
For two spin-0 particles, the only part of the wavefunction that changes under exchange is the spatial part, which carries a factor $(-1)^L$. Bose symmetry requires:

$$(-1)^L = +1 \quad \Longrightarrow \quad L = \text{even} \quad (L = 0, 2, 4, \ldots)$$

But we already showed:

- Angular momentum conservation requires $L = 0$
- Parity conservation requires $L = \text{odd}$
- Bose symmetry requires $L = \text{even}$

All three constraints cannot be simultaneously satisfied. The $\pi^0\pi^0$ channel is **triply forbidden**.

2.4 Summary: Three Independent Reasons

The decay $\eta \to 2\pi$ is forbidden by both the strong and electromagnetic interactions for **three independent reasons**:

1. Parity + Angular Momentum: The η has $J^P = 0^-$. Angular momentum conservation forces $L = 0$ for the two-pion system, giving $P_{\text{final}} = (-1)^0 = +1$. But $P_\eta = -1$. Parity is violated.

2. CP Conservation: The η has $CP = -1$, but the $\pi^+\pi^-$ system has $CP = +1$ for *every* value of L. No orbital angular momentum can fix this.

3. Bose Symmetry (for $\pi^0\pi^0$): Two identical π^0's require $L = \text{even}$. Parity requires $L = \text{odd}$. Angular momentum requires $L = 0$. All three are mutually contradictory.

Since both strong and electromagnetic interactions conserve P, C, and CP, the 2π decay is forbidden by both.

3 Part (b): Why is $\eta \to 3\pi$ Forbidden Strongly but Allowed Electromagnetically?

Now the situation changes. We need to find a symmetry that:

- Is conserved by the strong interaction (forbidding the decay)
- Is *not* conserved by the electromagnetic interaction (allowing the decay)

The answer is G**-parity** (equivalently, **isospin**).

3.1 The G-Parity Argument

Step 1: G-parity of the η.
We computed this in Problem 4.36:

$$G(\eta) = (-1)^I \cdot C = (-1)^0 \cdot (+1) = +1$$

Step 2: G-parity of n pions.
Each pion has $G(\pi) = -1$. Since G-parity is multiplicative:

$$G(3\pi) = (-1)^3 = -1$$

Step 3: Compare.

$$G(\eta) = +1 \qquad \neq \qquad G(3\pi) = -1$$

G-parity is violated. The strong interaction conserves G-parity, so it **cannot** mediate $\eta \to 3\pi$.

The η has $G = +1$, but three pions have $G = (-1)^3 = -1$. Since G-parity is conserved in the strong interaction:

$$G_{\text{initial}} \neq G_{\text{final}} \quad \Longrightarrow \quad \text{Decay forbidden by strong interaction}$$

3.2 Why the Electromagnetic Interaction Allows It

G-parity is defined as $G = C \cdot e^{i\pi I_2}$, which involves an **isospin rotation**. The electromagnetic interaction **does not conserve isospin** (because it distinguishes between up quarks with charge $+\frac{2}{3}$ and down quarks with charge $-\frac{1}{3}$).
Since G-parity is built on isospin, and the electromagnetic interaction breaks isospin, **the electromagnetic interaction does not conserve G-parity**. Therefore, the G-parity selection rule does not apply, and $\eta \to 3\pi$ is allowed electromagnetically.

Let's also verify that no other symmetry blocks it:

- **Parity:** $P(\eta) = -1$. For 3π with $L_1 = L_2 = 0$: $P(3\pi) = (-1)^3 = -1$. ✓ Parity conserved.
- **Angular momentum:** $J(\eta) = 0$ can be achieved with $L_1 = L_2 = 0$. ✓
- **C-parity:** $C(\eta) = +1$. For the $3\pi^0$ channel, $C(3\pi^0) = (+1)^3 = +1$. ✓
- **CP:** $CP(\eta) = -1$, and $CP(3\pi) = -1$ (since $P = -1$ and $C = +1$). ✓
- **Charge:** Conserved in all channels. ✓

No conserved symmetry of the electromagnetic interaction is violated, so the decay is **allowed**.

3.3 Equivalent Argument: Direct Isospin Analysis

The G-parity argument can be restated purely in terms of isospin, which some find more transparent.

The η has $I = 0$. Each pion has $I = 1$. If the strong interaction (which conserves isospin) were to mediate $\eta \to 3\pi$, the total isospin of the three-pion system would need to be $I = 0$.
But can three pions in an s-wave ($L_1 = L_2 = 0$) form an $I = 0$ state? The answer is **no**, for a subtle reason involving Bose symmetry.
Three identical spin-0 bosons (pions) in s-wave must have a **totally symmetric** spatial wavefunction. The isospin wavefunction must therefore also be **totally symmetric** (since the total wavefunction = spatial × isospin must be symmetric for bosons).
However, the $I = 0$ state formed from three $I = 1$ particles is **totally antisymmetric** under exchange. This contradicts the Bose symmetry requirement. Therefore:

There is no way to construct a three-pion state with $I = 0$ and the correct Bose symmetry.

This is the isospin version of the same statement that G-parity makes in one line.

The G-parity argument and the isospin argument are not two different physics arguments — they are the **same argument** expressed in two different languages. G-parity was specifically invented by Lee and Yang in 1956 to *encode* isospin selection rules into a single multiplicative quantum number, making them as easy to check as parity conservation.
The rule $G(n\pi) = (-1)^n$ is a compact restatement of what would otherwise require a detailed Bose symmetry + Clebsch–Gordan analysis.

4 The Deeper Story: What Actually Drives $\eta \to 3\pi$?

It was initially assumed that $\eta \to 3\pi$ was a purely electromagnetic decay (since EM breaks isospin). However, in 1966–67, Sutherland and Veltman proved that the **electromagnetic amplitude for $\eta \to 3\pi$ actually vanishes** in the limit of zero pion mass. This is the famous "Sutherland theorem."
The true driving mechanism is the **up–down quark mass difference**: $m_d - m_u \approx 2.5$ MeV. This mass difference is a term in the QCD Lagrangian that breaks isospin

symmetry *within the strong sector itself*. It enables η–π^0 mixing, which opens the 3π channel.
The decay amplitude is proportional to $(m_d - m_u)$, making $\eta \to 3\pi$ one of the most important processes for determining quark mass ratios. This is why Griffiths calls it an electromagnetic decay (it violates isospin), but the modern understanding is more nuanced: it's driven by the quark mass difference, with small electromagnetic corrections.

5 Why the η is "Stable"

The η meson at 549 MeV/c^2 has more than enough mass to decay to 2π (≈ 280 MeV) or 3π (≈ 420 MeV). Yet:

Decay	Strong?	EM?
$\eta \to 2\pi$	**Forbidden** (Parity, CP)	**Forbidden** (Parity, CP)
$\eta \to 3\pi$	**Forbidden** (G-parity)	**Allowed**
$\eta \to 2\gamma$	—	**Allowed**
$\eta \to \pi\pi\gamma$	—	**Allowed**

Every strong decay mode is forbidden. The η can only decay through the electromagnetic interaction (or the quark mass difference). Electromagnetic decays are much slower than strong decays ($\alpha_{\text{EM}} \approx 1/137$ vs $\alpha_s \approx 1$), giving the η a total width of only $\Gamma \approx 1.3$ keV — **five orders of magnitude narrower** than a typical strong resonance. This is why it is classified as "stable" in particle physics.

6 Summary of Key Arguments

Part (a): $\eta \to 2\pi$ is forbidden by both strong and electromagnetic interactions because:

1. **Parity contradiction:** Angular momentum conservation forces $L = 0$, but parity conservation requires $L =$ odd. These are incompatible.
2. **CP violation:** $CP(\eta) = -1$ but $CP(\pi^+\pi^-) = +1$ for all values of L. No escape.
3. **Bose symmetry** (for $\pi^0\pi^0$): Identical bosons require $L =$ even, parity requires $L =$ odd, J conservation requires $L = 0$. Triply impossible.

Part (b): $\eta \to 3\pi$ is forbidden as a strong decay but allowed electromagnetically because:

1. The η has $G = +1$, but $G(3\pi) = (-1)^3 = -1$. G-parity is violated.
2. The strong interaction conserves G-parity $\Rightarrow$ decay **forbidden** strongly.
3. The electromagnetic interaction does **not** conserve G-parity (because it breaks isospin) $\Rightarrow$ decay **allowed** electromagnetically.
4. All symmetries conserved by EM (P, C, CP, charge) are satisfied in $\eta \to 3\pi$.

7 Appendix: Observed η Decay Modes

Decay	BR (%)	Mechanism	Interaction
$\eta \to \gamma\gamma$	39.4	Chiral anomaly	EM
$\eta \to 3\pi^0$	32.6	$m_d - m_u$	Isospin violation
$\eta \to \pi^+\pi^-\pi^0$	22.9	$m_d - m_u$	Isospin violation
$\eta \to \pi^+\pi^-\gamma$	4.2	WZW anomaly	EM
$\eta \to e^+e^-\gamma$	0.7	Dalitz decay	EM
$\eta \to \pi^+\pi^-$	$< 1.6 \times 10^{-5}$	—	Forbidden (P, CP)
$\eta \to \pi^0\pi^0$	$< 3.5 \times 10^{-4}$	—	Forbidden (P, CP, Bose)

The absence of strong decays and the dominance of EM/isospin-violating modes is exactly what the symmetry arguments predict.